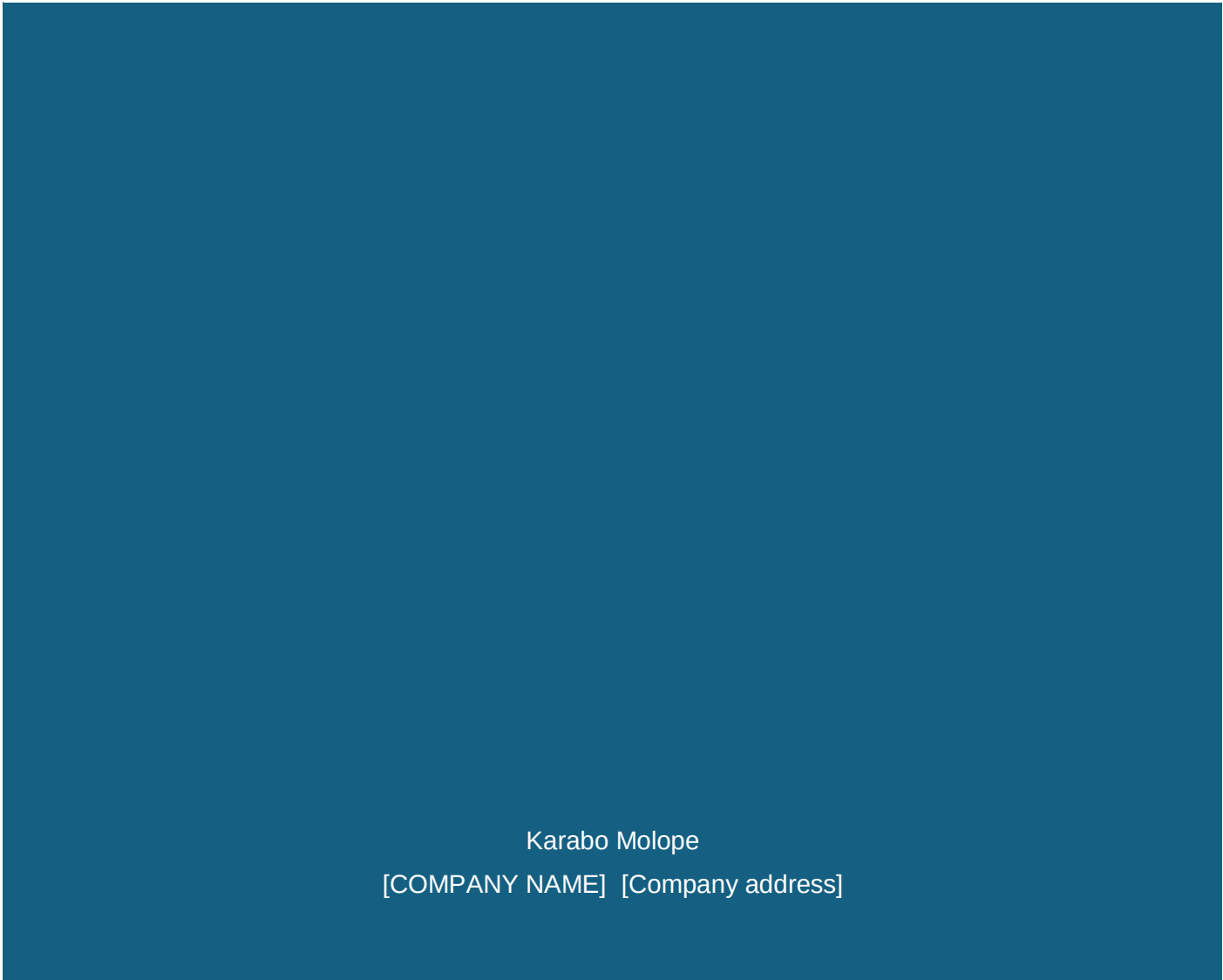




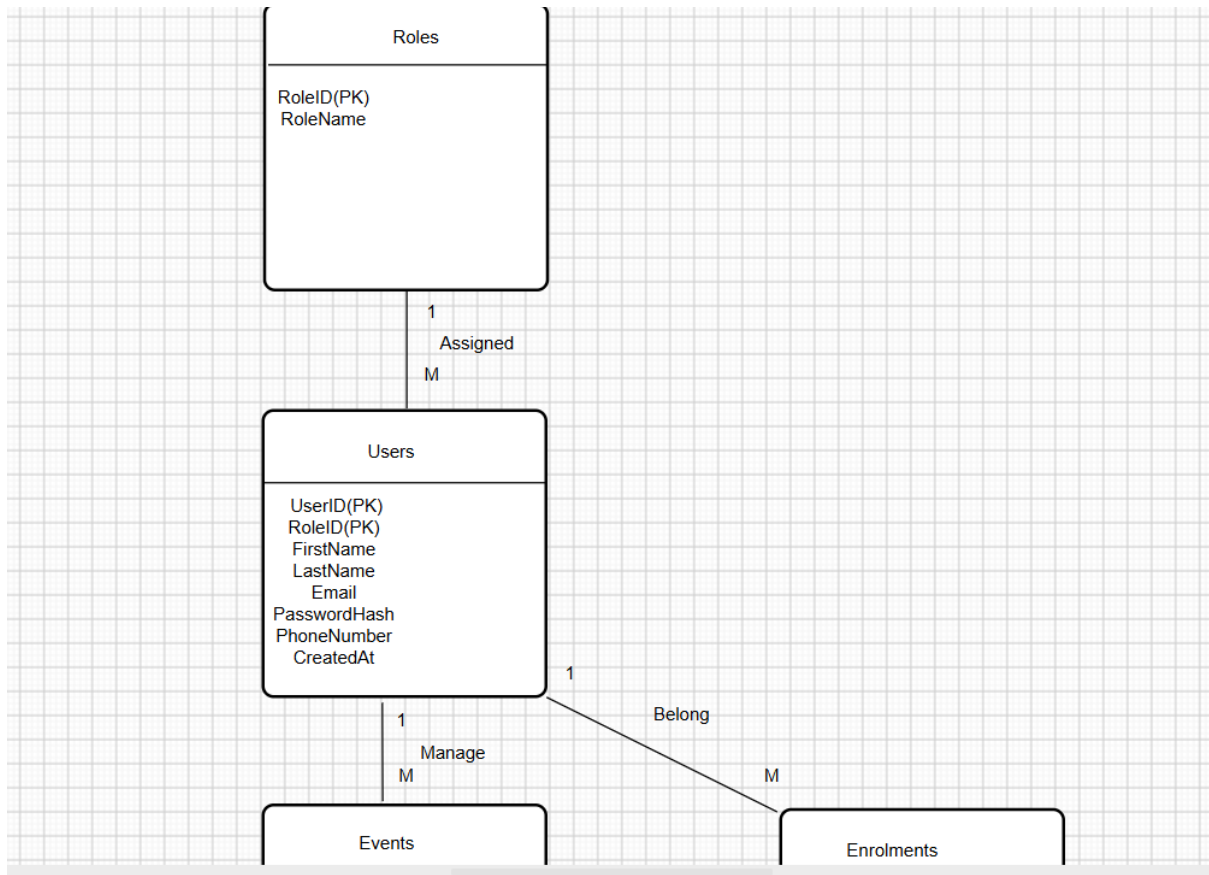
ST10484258 KARABO MOLOPE PROG6212 PART 1 GROUP 4

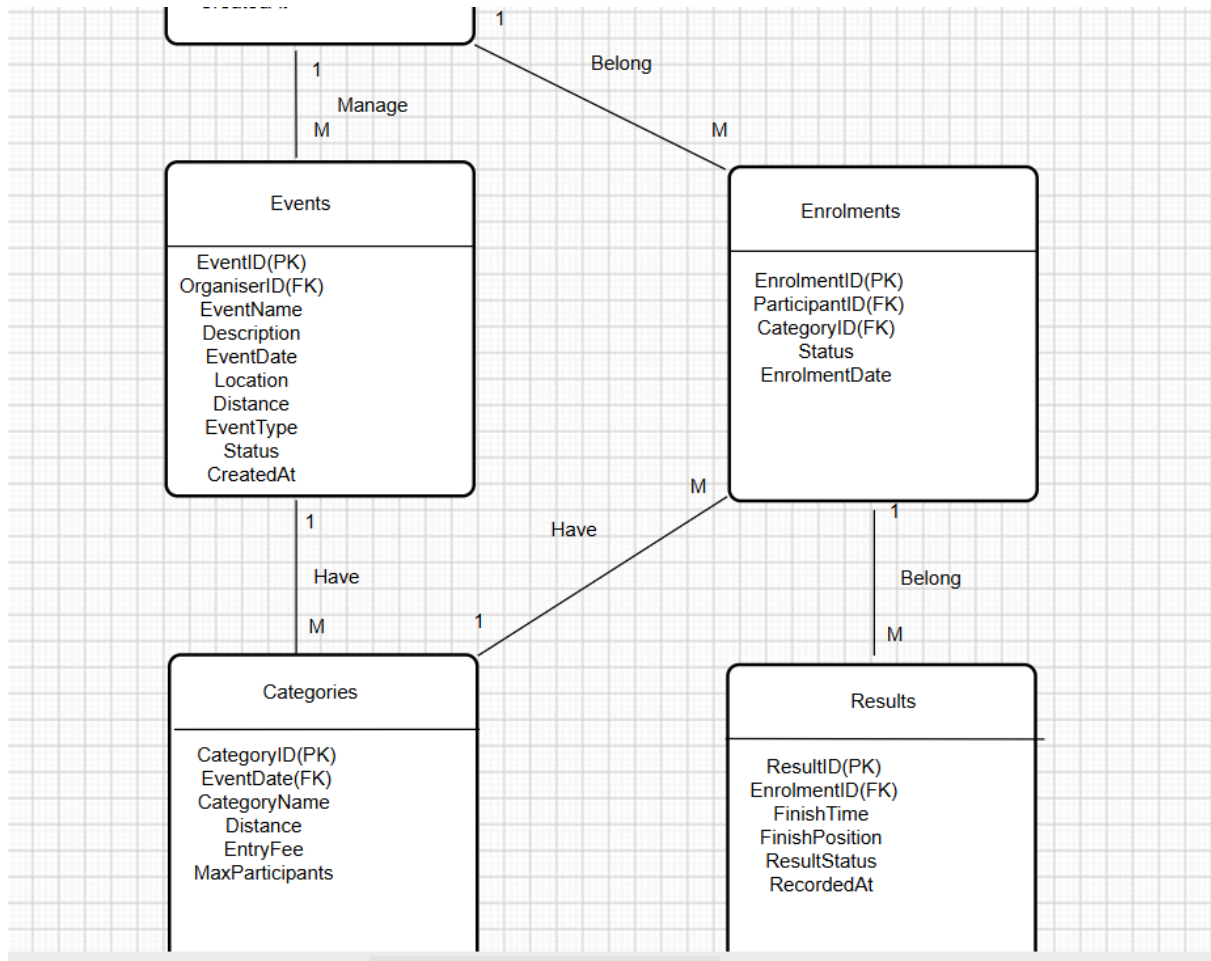


Karabo Molope
[COMPANY NAME] [Company address]

Section A

Entity Relationship Diagram





SECTION B

Authentication Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
Post	/api/auth/register	Creates a new Participant account in the RaceDay system.	None	{ firstName, lastName, email, password, phoneNumber }	201 Created – newly created user. 400 Bad Request – invalid data or email already exists.
Post	/api/auth/login	Authenticates a registered user and returns an authentication token.	None	{ email, password }	200 OK – authentication token and user information. 401 Unauthorized – invalid credentials.

User Profile Endpoint

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
Get	/api/users/me	Retrieves the profile information of the currently authenticated user.	Any authenticated user	None	200 OK – user's profile information. 401 Unauthorized – user is not logged in.
Get	/api/users/me	Updates the profile information of the currently authenticated user.	Any authenticated user	{ firstName, lastName, phoneNumber }	200 OK – updated profile. 400 Bad Request – invalid information. 401 Unauthorized – user is not logged in.

Event Endpoint

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
Get	/api/events	Retrieves a list of available RaceDay events for participants to browse.	None	None	200 OK – list of events.
Get	/api/events/{id}	Retrieves the details of a specific event.	None	None	200 OK – event details. 404 Not Found – event does not exist.
Post	/api/events	Creates a new RaceDay event.	Organiser	{ eventName, description, eventDate, location, distance, eventType, status }	201 Created – newly created event. 400 Bad Request – invalid event information. 401/403 – authentication or role failure.
Put	/api/events/{id}	Updates an existing event managed by the Organiser.	Organiser	{ eventName, description, eventDate, location, distance, eventType, status }	200 OK – updated event. 404 Not Found – event does not exist. 403 Forbidden – user is not authorised.
Delete	/api/events/{id}	Deletes an existing event managed by the Organiser.	Organiser	None	204 No Content – event successfully deleted. 404 Not Found – event does not exist. 403 Forbidden – user is not authorised.

Category Endpoint

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
Get	/api/events/{eventId}/categories	Retrieves all categories belonging to a specific event.	None	None	200 OK – list of categories. 404 Not Found – event does not exist.
Get	/api/categories/{id}		None	None	200 OK – category details. 404 Not Found – category does not exist.
Post	/api/events/{eventId}/categories	Creates a new category for an event.	Organiser	Creates a new category for an event.	201 Created – newly created category. 400 Bad Request – invalid data.
Put	/api/categories/{id}	Updates an existing event category.	Organiser	{ categoryName, distance, entryFee, maximumParticipants }	200 OK – updated category. 404 Not Found – category does not exist. 403 Forbidden – user is not authorised.

Delete	/api/categories/{id}	Deletes an existing event category.	Organiser	None	204 No Content – category deleted. 404 Not Found – category does not exist. 409 Conflict – category cannot be deleted because it has enrolments.
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Event Enrolment Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
Get	/api/events/{eventId}/categories	Retrieves all categories belonging to a specific event.	None	None	201 Created – enrolment created. 404 Not Found – category does not exist. 409 Conflict – participant is already enrolled or category is full.

Get	/api/categories/{id}	Retrieves the details of a specific category .	None	None	200 OK – category details. 404 Not Found – category does not exist.
Post	/api/events/{eventId}/categories	Creates a new category for an event.	Organiser	{ categoryName, distance, entryFee, maximumParticipants }	201 Created – newly created category. 400 Bad Request – invalid data.
Put	/api/categories/{id}	Updates an existing event category .	Organiser	{ categoryName, distance, entryFee, maximumParticipants }	200 OK – updated category. 404 Not Found – category does not exist. 403 Forbidden – user is not authorised.
Delete	/api/categories/{id}	Deletes an existing event category .	Organiser	None	204 No Content – category deleted. 404 Not Found – category does not exist. 409 Conflict – category cannot be

					deleted because it has enrolments.
--	--	--	--	--	------------------------------------

Result Endpoints

HTTP method	Route	Description	Role Required	Request Body	Expected Response
Post	/api/enrolments/{enrolmentId}/results	Records a participant's result for a completed enrolment.	Organiser	{ finishTime, finishPosition, resultStatus }	201 Created – result recorded. 404 Not Found – enrolment does not exist. 409 Conflict – result already exists
Put	/api/results/{id}	Updates an existing participant result.	Organiser	{ finishTime, finishPosition, resultStatus }	200 OK – updated result. 404 Not Found – result does not exist. 403 Forbidden – user is not authorised.
Get	/api/results/me	Retrieves the results belonging to the currently authenticated Participant.	Participant	None	200 OK – participant's personal results. 401 Unauthorized – user is not

					logged in.
Get	/api/events/{eventId}/results	Retrieves the results for participants who completed a specific event.	Organiser	None	200 OK – event results. 404 Not Found – event does not exist. 403 Forbidden – user is not authorised.
Get	/api/results/{id}	Retrieves a specific participant result.	Participant/Organiser	None	200 OK – event results. 404 Not Found – result does not exist. 403 Forbidden – user is not authorised.

Complete API Endpoint Summary

Area	Method	Endpoint	Role
Authentication	Post	/api/auth/register	None
Authentication	Post	/api/auth/login	None
User Profile	Get	/api/users/me	Authenticated
User Profile	Put	/api/users/me	Authenticated
Events	Get	/api/events	Public
Events	Get	/api/events/{id}	Public
Events	Post	/api/events	Organiser
Events	Put	/api/events/{id}	Organiser
Events	Delete	/api/events/{id}	Organiser
Categories	Get	/api/events/{eventId}/categories	Public
Categories	Get	/api/categories/{id}	Public
Categories	Post	/api/events/{eventId}/categories	Organiser
Categories	Put	/api/categories/{id}	Organiser
Categories	Delete	/api/categories/{id}	Organiser

Enrolments	Post	/api/categories/{categoryId}/enrolments	Participant
Enrolments	Get	/api/enrolments/me	Participant
Enrolments	Get	/api/events/{eventId}/enrolments	Organiser
Enrolments	Get	/api/enrolments/{id}	Participant/Organiser
Results	Post	/api/enrolments/{enrolmentId}/results	Organiser
Results	Put	/api/results/{id}	Organiser
Results	Get	/api/results/me	Participant
Results	Get	/api/events/{eventId}/results	Organiser
Results	Get	/api/results/{id}	Participant/Organiser

SECTION C

SQL

IF DB_ID('RaceDayDB') IS NULL

BEGIN

 CREATE DATABASE RaceDayDB;

END;

USE RaceDayDB;

IF OBJECT_ID('dbo.Results', 'U') IS NOT NULL

 DROP TABLE dbo.Results;

IF OBJECT_ID('dbo.Enrolments', 'U') IS NOT NULL

 DROP TABLE dbo.Enrolments;

IF OBJECT_ID('dbo.Categories', 'U') IS NOT NULL

 DROP TABLE dbo.Categories;

IF OBJECT_ID('dbo.Events', 'U') IS NOT NULL

 DROP TABLE dbo.Events;

IF OBJECT_ID('dbo.Users', 'U') IS NOT NULL

 DROP TABLE dbo.Users;

IF OBJECT_ID('dbo.Roles', 'U') IS NOT NULL

 DROP TABLE dbo.Roles;

CREATE TABLE dbo.Roles

(

 RoleID INT IDENTITY(1,1) NOT NULL,

 RoleName VARCHAR(50) NOT NULL,

CONSTRAINT PK_Roles

 PRIMARY KEY (RoleID),

```
CONSTRAINT UQ_Roles_RoleName
    UNIQUE (RoleName)
);
CREATE TABLE dbo.Users
(
    UserID INT IDENTITY(1,1) NOT NULL,
    RoleID INT NOT NULL,
    FirstName VARCHAR(50) NOT NULL,
    LastName VARCHAR(50) NOT NULL,
    Email VARCHAR(150) NOT NULL,
    PasswordHash VARCHAR(255) NOT NULL,
    PhoneNumber VARCHAR(20) NULL,
    CreatedAt DATETIME2 NOT NULL
    CONSTRAINT DF_Users_CreatedAt
        DEFAULT SYSDATETIME(),

    CONSTRAINT PK_Users
        PRIMARY KEY (UserID),

    CONSTRAINT UQ_Users_Email
        UNIQUE (Email),

    CONSTRAINT FK_Users_Roles
        FOREIGN KEY (RoleID)
            REFERENCES dbo.Roles(RoleID)
);
CREATE TABLE dbo.Events
(
    EventID INT IDENTITY(1,1) NOT NULL,
```

```
OrganiserID INT NOT NULL,  
EventName VARCHAR(150) NOT NULL,  
Description VARCHAR(500) NULL,  
EventDate DATE NOT NULL,  
Location VARCHAR(200) NOT NULL,  
Distance DECIMAL(6,2) NOT NULL,  
EventType VARCHAR(30) NOT NULL,  
Status VARCHAR(30) NOT NULL  
    CONSTRAINT DF_Events_Status  
    DEFAULT 'Upcoming',  
CreatedAt DATETIME2 NOT NULL  
    CONSTRAINT DF_Events_CreatedAt  
    DEFAULT SYSDATETIME(),  
  
CONSTRAINT PK_Events  
    PRIMARY KEY (EventID),  
  
CONSTRAINT FK_Events_Organiser  
    FOREIGN KEY (OrganiserID)  
    REFERENCES dbo.Users(UserID),  
  
CONSTRAINT CK_Events_Distance  
    CHECK (Distance > 0),  
  
CONSTRAINT CK_Events_EventType  
    CHECK (EventType IN ('Running', 'Walking', 'Cycling')),  
  
CONSTRAINT CK_Events_Status  
    CHECK (Status IN ('Upcoming', 'Completed', 'Cancelled'))
```

```
);  
  
CREATE TABLE dbo.Categories  
(  
    CategoryID INT IDENTITY(1,1) NOT NULL,  
    EventID INT NOT NULL,  
    CategoryName VARCHAR(100) NOT NULL,  
    Distance DECIMAL(6,2) NOT NULL,  
    EntryFee DECIMAL(10,2) NOT NULL,  
    MaximumParticipants INT NOT NULL,  
  
    CONSTRAINT PK_Categories  
        PRIMARY KEY (CategoryID),  
  
    CONSTRAINT FK_Categories_Events  
        FOREIGN KEY (EventID)  
        REFERENCES dbo.Events(EventID),  
  
    CONSTRAINT UQ_Categories_Event_Category  
        UNIQUE (EventID, CategoryName),  
  
    CONSTRAINT CK_Categories_Distance  
        CHECK (Distance > 0),  
  
    CONSTRAINT CK_Categories_EntryFee  
        CHECK (EntryFee >= 0),  
  
    CONSTRAINT CK_Categories_MaxParticipants  
        CHECK (MaximumParticipants > 0)  
);
```

```
CREATE TABLE dbo.Enrolments
(
    EnrolmentID INT IDENTITY(1,1) NOT NULL,
    ParticipantID INT NOT NULL,
    CategoryID INT NOT NULL,
    EnrolmentDate DATETIME2 NOT NULL
        CONSTRAINT DF_Enrolments_EnrolmentDate
        DEFAULT SYSDATETIME(),
    Status VARCHAR(30) NOT NULL
        CONSTRAINT DF_Enrolments_Status
        DEFAULT 'Confirmed',

    CONSTRAINT PK_Enrolments
        PRIMARY KEY (EnrolmentID),

    CONSTRAINT FK_Enrolments_Participant
        FOREIGN KEY (ParticipantID)
        REFERENCES dbo.Users(UserID),

    CONSTRAINT FK_Enrolments_Category
        FOREIGN KEY (CategoryID)
        REFERENCES dbo.Categories(CategoryID),

    CONSTRAINT UQ_Enrolments_Participant_Category
        UNIQUE (ParticipantID, CategoryID),

    CONSTRAINT CK_Enrolments_Status
        CHECK (Status IN ('Pending', 'Confirmed', 'Cancelled'))
);
```

```
CREATE TABLE dbo.Results
(
    ResultID INT IDENTITY(1,1) NOT NULL,
    EnrolmentID INT NOT NULL,
    FinishTime TIME NOT NULL,
    FinishPosition INT NOT NULL,
    ResultStatus VARCHAR(30) NOT NULL
        CONSTRAINT DF_Results_ResultStatus
        DEFAULT 'Official',
    RecordedAt DATETIME2 NOT NULL
        CONSTRAINT DF_Results_RecordedAt
        DEFAULT SYSDATETIME(),

    CONSTRAINT PK_Results
        PRIMARY KEY (ResultID),

    CONSTRAINT FK_Results_Enrolments
        FOREIGN KEY (EnrolmentID)
        REFERENCES dbo.Enrolments(EnrolmentID),

    CONSTRAINT UQ_Results_Enrolment
        UNIQUE (EnrolmentID),

    CONSTRAINT CK_Results_FinishPosition
        CHECK (FinishPosition > 0),

    CONSTRAINT CK_Results_Status
        CHECK (ResultStatus IN ('Official', 'Pending', 'Disqualified'))
);
```

```
INSERT INTO dbo.Roles
```

```
(  
    RoleName  
)
```

```
VALUES
```

```
    ('Organiser'),  
    ('Participant');
```

```
INSERT INTO dbo.Users
```

```
(  
    RoleID,  
    FirstName,  
    LastName,  
    Email,  
    PasswordHash,  
    PhoneNumber  
)
```

```
VALUES
```

```
(  
    1,  
    'Thabo',  
    'Mokoena',  
    'thabo.mokoena@raceday.co.za',  
    'HASHED_PASSWORD_ORGANISER_001',  
    '0712345678'
```

```
),
```

```
(  
    1,  
    'Lerato',  
    'Molefe',
```

```
'lerato.molefe@raceday.co.za',  
'HASHED_PASSWORD_ORGANISER_002',  
'0723456789'  
,  
(  
    2,  
    'Karabo',  
    'Molepo',  
    'karabo.molepo@example.com',  
    'HASHED_PASSWORD_PARTICIPANT_001',  
    '0734567890'  
,  
(  
    2,  
    'Naledi',  
    'Dlamini',  
    'naledi.dlamini@example.com',  
    'HASHED_PASSWORD_PARTICIPANT_002',  
    '0745678901'  
);
```

```
INSERT INTO dbo.Events
```

```
(  
    OrganiserID,  
    EventName,  
    Description,  
    EventDate,  
    Location,  
    Distance,  
    EventType,
```

Status

)

VALUES

(

1,

'Johannesburg City Run',

'Annual road-running event through Johannesburg.',

'2026-10-18',

'Johannesburg, Gauteng',

21.10,

'Running',

'Upcoming'

),

(

1,

'Pretoria Charity Walk',

'Community walking event supporting local charities.',

'2026-11-07',

'Pretoria, Gauteng',

10.00,

'Walking',

'Upcoming'

),

(

2,

'Soweto Cycling Challenge',

'Cycling event suitable for recreational and experienced cyclists.',

'2026-11-21',

'Soweto, Gauteng',

```

40.00,
'Cycling',
'Upcoming'
);
INSERT INTO dbo.Categories
(
    EventID,
    CategoryName,
    Distance,
    EntryFee,
    MaximumParticipants
)
VALUES
/* Johannesburg City Run */
(
    1,
    '5 KM Fun Run',
    5.00,
    80.00,
    500
),
(
    1,
    '10 KM Road Race',
    10.00,
    120.00,
    750
),
(

```

```
1,  
'21.1 KM Half Marathon',  
21.10,  
180.00,  
1000  
,
```

```
/* Pretoria Charity Walk */
```

```
(  
2,  
'5 KM Charity Walk',  
5.00,  
50.00,  
400  
,
```

```
(  
2,  
'10 KM Charity Walk',  
10.00,  
80.00,  
600  
,
```

```
/* Soweto Cycling Challenge */
```

```
(  
3,  
'20 KM Recreational Ride',  
20.00,  
100.00,
```

```

300
),
(
3,
'40 KM Cycling Challenge',
40.00,
160.00,
500
);
INSERT INTO dbo.Enrolments
(
ParticipantID,
CategoryID,
EnrolmentDate,
Status
)
VALUES
(
3,
1,
'2026-09-01 09:30:00',
'Confirmed'
),
(
3,
4,
'2026-09-02 10:15:00',
'Confirmed'
),

```

```
(
    4,
    2,
    '2026-09-02 11:00:00',
    'Confirmed'
),
(
    4,
    7,
    '2026-09-03 14:20:00',
    'Confirmed'
);
```

```
INSERT INTO dbo.Results
```

```
(
    EnrolmentID,
    FinishTime,
    FinishPosition,
    ResultStatus
```

```
)
VALUES
```

```
(
    1,
    '00:28:45',
    15,
    'Official'
```

```
),
(
    2,
    '00:42:18',
```

```
27,  
'Official'  
);
```

```
SELECT *  
FROM dbo.Roles;
```

100 % No issues found

Results Messages

	RoleID	RoleName
1	1	Organiser
2	2	Participant

Query executed successfully.

```
SELECT *  
FROM dbo.Users;
```

100 % No issues found Ln: 503, Ch: 16 (25 chars, 2 lines) | 5

Results Messages

	UserID	RoleID	FirstName	LastName	Email	PasswordHash	PhoneNumber	CreatedAt
1	1	1	Thabo	Mokoena	thabo.mokoena@raceday.co.za	HASHED_PASSWORD_ORGANISER_001	0712345678	2026-09-04 19:05:47.6173490
2	2	1	Lerato	Molefe	lerato.molefe@raceday.co.za	HASHED_PASSWORD_ORGANISER_002	0723456789	2026-09-04 19:05:47.6173490
3	3	2	Karabo	Molepo	karabo.molepo@example.com	HASHED_PASSWORD_PARTICIPANT_001	0734567890	2026-09-04 19:05:47.6173490
4	4	2	Naledi	Dlamini	naledi.dlamini@example.com	HASHED_PASSWORD_PARTICIPANT_002	0745678901	2026-09-04 19:05:47.6173490

Query executed successfully. (localdb)\api_both (15.0 RTM) Lab\VM2049939\Student (51) RaceDayDB 00:00:00

```
SELECT *  
FROM dbo.Events;
```

Results Messages										
	EventID	OrganiserID	EventName	Description	EventDate	Location	Distance	EventType	Status	CreatedAt
1	1	1	Johannesburg City Run	Annual road-running event through Johannesburg.	2026-10-18	Johannesburg, Gauteng	21.10	Running	Upcoming	2026-09-04 19:06:34.1258093
2	2	1	Pretoria Charity Walk	Community walking event supporting local chariti...	2026-11-07	Pretoria, Gauteng	10.00	Walking	Upcoming	2026-09-04 19:06:34.1258093
3	3	2	Soweto Cycling Challenge	Cycling event suitable for recreational and experi...	2026-11-21	Soweto, Gauteng	40.00	Cycling	Upcoming	2026-09-04 19:06:34.1258093

```
SELECT *
FROM dbo.Categories;
```

Results Messages						
	CategoryID	EventID	CategoryName	Distance	EntryFee	MaximumParticipants
1	1	1	5 KM Fun Run	5.00	80.00	500
2	2	1	10 KM Road Race	10.00	120.00	750
3	3	1	21.1 KM Half Marathon	21.10	180.00	1000
4	4	2	5 KM Charity Walk	5.00	50.00	400
5	5	2	10 KM Charity Walk	10.00	80.00	600
6	6	3	20 KM Recreational Ride	20.00	100.00	300
7	7	3	40 KM Cycling Challenge	40.00	160.00	500

Query executed successfully. (localdb)\api_both (15.0 RTM) LabV

```
SELECT *
FROM dbo.Enrolments;
```

Results Messages					
	EnrolmentID	ParticipantID	CategoryID	EnrolmentDate	Status
1	1	3	1	2026-09-01 09:30:00.0000000	Confirmed
2	2	3	4	2026-09-02 10:15:00.0000000	Confirmed
3	3	4	2	2026-09-02 11:00:00.0000000	Confirmed
4	4	4	7	2026-09-03 14:20:00.0000000	Confirmed

Query executed successfully. (localdb)\api_bot

```
SELECT *
FROM dbo.Results;
```

Results

Messages

	ResultID	EnrolmentID	FinishTime	FinishPosition	ResultStatus	RecordedAt
1	1	1	00:28:45.0000000	15	Official	2026-09-04 19:10:10.6009752
2	2	2	00:42:18.0000000	27	Official	2026-09-04 19:10:10.6009752

Query executed successfully.

(localdb)\api_both (15.0 R

SELECT

u.FirstName + ' ' + u.LastName AS Participant,
e.EventName,
c.CategoryName,
en.Status AS EnrolmentStatus,
r.FinishTime,
r.FinishPosition,
r.ResultStatus

FROM dbo.Enrolments en

INNER JOIN dbo.Users u

ON en.ParticipantID = u.UserID

INNER JOIN dbo.Categories c

ON en.CategoryID = c.CategoryID

INNER JOIN dbo.Events e

ON c.EventID = e.EventID

LEFT JOIN dbo.Results r

ON en.EnrolmentID = r.EnrolmentID;

Results Messages

	Participant	EventName	CategoryName	EnrolmentStatus	FinishTime	FinishPosition	ResultStatus
1	Karabo Molepo	Johannesburg City Run	5 KM Fun Run	Confirmed	00:28:45.0000000	15	Official
2	Karabo Molepo	Pretoria Charity Walk	5 KM Charity Walk	Confirmed	00:42:18.0000000	27	Official
3	Naledi Dlamini	Johannesburg City Run	10 KM Road Race	Confirmed	NULL	NULL	NULL
4	Naledi Dlamini	Soweto Cycling Challenge	40 KM Cycling Challenge	Confirmed	NULL	NULL	NULL

Query executed successfully.

(localdb)\api_both (15.0 RTM) LabVM2049939\Stud